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SUPERSEDING
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MILITARY SPECIFICATION

CRUSHING AND SCREENING PLANT: TWO-UNIT, DIESEL-ENGINE-DRIVEN, SEMITRAILER-MOUNTED, 35-TON-PER-HOUR CAPACITY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a two-unit crushing and screening plant consisting of a jaw and a roll-type crusher with a capacity of 35 tons per hour when operating as a plant. Each unit is diesel-engine driven, is mounted separately on a semitrailer, and is furnished with a dolly converter.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-B-171	-	Bearings, Ball, Annular (General Purpose).
FF-B-185	-	Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	-	Bearing, Roller, Tapered.
QQ-S-681	-	Steel Castings.
QQ-S-781	-	Steel, Strapping, Flat.
QQ-S-790	-	Steel Strapping, Round (Bare and Zinc-Coated).
SS-R-406	-	Road and Paving Materials; Methods of Sampling and Testing.
TT-P-664	-	Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-121	-	Varnish, Spar, Water-Resisting.
UU-P-271	-	Paper, Wrapping, Waterproofed Kraft.
UU-T-81	-	Tags, Shipping and Stock.
ZZ-B-206	-	Belting, Flat; Conveyor, Rubber.
ZZ-R-765	-	Rubber, Silicone: Low- and High-Temperature and Tear Resistant.
GGG-J-51	-	Jacks, Mechanical and Hydraulic, Hand Operated.
KKK-L-370	-	Lining, Friction, (Clutch and Brake; Metallic, Metal-Ceramic and Semimetallic).
MMM-A-260	-	Adhesive, Water-Resistant, (for Sealing Waterproofed Paper).
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.

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PPP-C-650 - Crates, Wood, Open and Covered.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

Military

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-130 - Paper, Wrapping, Laminated and Creped.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514 - Plates, Identification; Transportation-Data and Blank.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-F-3541 - Fittings, Lubrication.
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-G-3859 - Grease Guns, Hand, High-Pressure, Lever-Operated (15-Ounce and 21-Ounce Capacity).
MIL-H-3992 - Hose and Hose Assembly, Rubber; Air and Vacuum Brake, Automotive.
MIL-S-8660 - Silicone Compound.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-E-11276 - Engines, Diesel: Industrial, Medium- and High-Speed, General Specification for.
MIL-R-12144 - Reflector, Indicating, Clearance.
MIL-C-13486 - Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty, Single Conductor and Multiconductor.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.
MIL-P-13983 - Paint, Temporary, Lusterless, Gasoline Removable.
MIL-I-15126 - Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-H-15424 - Hand Tools; Packaging of.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-E-16298 - Electric Machines having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-52058 - Chains, Roller, Power Transmission and Conveyor.
MIL-A-52363 - Air Cleaners, Intake: Dry-Type (for Internal Combustion Engine).
MIL-E-55301 - Electromagnetic Compatibility.

STANDARDS

Federal

FED. STD. NO. 595 - Colors.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-281	-	Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-424	-	Chain, Roller; Power Transmission, Offset Sidebars, Single-Pitch.
MIL-STD-642	-	Identification marking of Administrative, Combat, and Tactical Transport Vehicles.
MIL-STD-1400	-	Engines, Gasoline or Diesel, Methods of Test.
MS16961	-	Troughing Carrier Assembly, 4 In. Dia Roll, 4-5/8 In. Backing; 24 and 30 In. Belt Width.
MS16963	-	Troughing Carrier Assembly Impact, 4-In. Dia Rubber Roll, 4-5/8 In. Backing; 24- and 30-In. Belt Width.
MS 16964	-	Return Roll Assembly, 4 In. Dia Roll, for 4-5/8 In. Backing Trough Carrier; 24 and 30 In. Belt Width.
MS21322	-	Clamps, Loop - Aluminum - Plain and Cushioned.
MS21333	-	Clamps, Loop - Steel - Plain and Cushioned.
MS21334	-	Clamps, Loop - Double, Cable or Tube, Steel, Plain and Cushioned.
MS27142	-	Connector, Plug, Electrical - Pin Contact No. 12, 14 & 16 A. W. G., Waterproof.
MS27143	-	Connector, Plug, Electrical-Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	-	Connector, Plug, Electrical - Socket Contact, No. 14 & 16 A.W.G., Waterproof.
MS27145	-	Connector, Plug, Electrical - No. 14 A.W.G., Double Contact, Waterproof.
MS27146	-	Connector, Receptacle, Electrical - No. 12, 14 & 16 A.W.G., Double Contact, Waterproof.
MS27147	-	Adapter, Connector - 3 Way, No. 12, 14 & 16 A.W.G., Waterproof.
MS27148	-	Pin Contact, Electrical Connector - No. 12, 14 & 16 A.W.G.
MS33800	-	Connector, Receptacle, Electrical - Pin Contact, No. 12, 14 & 16 A.W.G., Waterproof.
MS35423	-	Light, Marker, Clearance - Service
MS35424	-	Light, Marker, Clearance - Blackout.
MS35643	-	Gasket: Cap, Fuel Tank.
MS35644	-	Filler Neck Assembly, Fuel Tank - Military Vehicles.
MS35645	-	Cap, Fuel Tank - Tactical Vehicles, Fording.
MS35647	-	Padlocks, Key Operated.
MS35746	-	Coupling, Automotive - Air Brake Lines.
MS35773	-	Radiators, Engine Cooling, Industrial.
MS35829	-	Hinge, Continuous (Piano), 0.075 Thickness, 0.187 and 0.250 Pin Dia.
MS35844	-	Plug, Machine Thread; Magnetic (Drain).
MS35865	-	Axle, Vehicular Non-Driving, Single, for Tactical Wheeled Vehicles.
MS39020	-	Band, Marker, Blank.
MS39062	-	Circuit Breakers: Thermal, 30 Volt D. C., Automatic, Waterproof, 15 to 30 Amp.
MS39127	-	Tray Screen Sizes and Edging for Aggregate Screens.
MS49006	-	Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.
MS51118	-	Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
MS51330	-	Stop Light - Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS51331	-	Guard, Splash, Wheel.
MS51337	-	Coupler, Drawbar, Ring - Heavy Duty.
MS53004	-	Valve, Relay, Emergency-Breakaway, Airbrake.
MS53007	-	Plate, Identification - Emergency and Service.

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MS53033	-	Fifth Wheel Assembly - 33 Inch Universal, Truck-Tractor, 25,000 Lbs. Capacity.
MS53036	-	Kingpin, Fifth Wheel, 2 Inch.
MS53063	-	Indicator, Air Cleaner, Intake; Restriction, Mechanical Type.
MS75021	-	Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.
MS500010	-	Truck, Dump, 5-Ton, 6 x 6, M-51.

DRAWINGS

MO(ATAC)

7056674	-	Cable, Waterproof, Electrical.
7056700	-	Terminal, Special Waterseal.
7056772	-	Insulator, Sleeve.
7070301	-	Chart, Electrical Circuit Numbers.
7723494	-	Electrical Connectors, Waterproof.
7728811	-	Assembly, Harness, Intervehiculer.
7731428	-	Cover Assy., Receptacle.
7969299	-	Clamp Assembly.
8347066	-	Schematic, Wiring, Typical, Dolly, 24 Volt, Waterproof.
8347201	-	Schematic Wiring Diagram, Semitype Trailer, (Typical) 24-Volt.
8668076	-	Waterproof Insulation for Electrical Components and Cable Splices.
8722728	-	Installation, Receptacle.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARDS ASSOCIATION

B1.5	-	Threads.
B6.6	-	Gear Tolerances and Inspection.
B55.1	-	Multiple V-Belt Drives.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A48	-	Gray Iron Castings.
A128	-	Austenitic Manganese-Steel Castings.
A193	-	Alloy-Steel Bolting Materials for High Temperature Services.
A242	-	High Strength Low Alloy Structural Steel.
A536	-	Ductile Iron Castings.
C170	-	Method of Test for Compressive Strength of Natural Building Stone.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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NATIONAL SAND AND GRAVEL ASSOCIATION

Circular No. 80, April 1959.

(Application for copies should be addressed to the National Sand and Gravel Association, 527 Munsey Building, Washington 4, D. C.)

INTERSTATE COMMERCE COMMISION

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N. Y. 10017.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Welding Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

RUBBER MANUFACTURERS ASSOCIATION, INC.

Engineering Standard, "Specifications for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections)."

(Applications for copies should be addressed to the Rubber Manufacturers Association, Inc., 444 Madison Avenue, New York, N. Y. 10022.)

AMERICAN TRUCKING ASSOCIATION

E-3 - Specification for Electrical Lighting and Wiring for Commercial Vehicle Bodies.

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(Application for copies should be addressed to the American Trucking Association, Incorporated, 1616 P Street NW, Washington, D. C. 20036.)

TIRE AND RIM ASSOCIATION, INC.

Tire and Rim Association Year Book.

(Application for copies should be addressed to the Tire and Rim Association, Inc., Command Building, 34 North Hawkins Avenue, Akron, Ohio.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The crushing and screening plant shall consist of two separate units, the primary unit, and the secondary unit. Each unit shall be mobile, self-contained, and equipped with a diesel engine as an integral part of the unit. Each unit shall be mounted on a semitrailer chassis that will permit towing either as a semitrailer or as a full trailer when used with a dolly converter. The plant shall operate as specified herein when set up with the longitudinal axes of the units in line or at an angle of 90 degrees to each other and with the primary unit feeding the secondary unit.

3.1.1 Capacity. When crushing quarry rock as specified in 3.1.1.1 which requires 80 percent crushing, the plant shall have a capacity of not less than 35 tons per hour passing 1-1/2 -inch square screen openings.

3.1.1.1 Quarry rock. The quarry rock shall be diabase or traprock having an average compressive strength of not less than 32,000 pounds per square inch (psi) when tested in accordance with ASTM C170 and shall have a percentage of wear of not more than 20 when tested in accordance with SS-R-406, method 208.11.

3.1.2 Stress limitations. The plant shall be constructed so that the maximum stress imposed on any structural component shall not exceed 60 percent of the yield point.

3.2 Preproduction model. The supplier shall furnish a plant within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods will produce plants that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish plants conforming to this specification.

3.3 Initial production. Unless otherwise specified (see 6.2), the supplier shall furnish to the Government one or more crushing and screening plant(s) (see 6.2) for testing as specified in 4.4.

3.4 Material. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. All materials shall be new and unused.

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3.5 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.5.1 ICC regulations. Each unit shall conform to all applicable requirements of the ICC Motor Carrier Safety Regulations, except as specified herein.

3.6 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates such as those on shrouds or housings which must be removed for adjustment or maintenance shall be equipped with quick-disconnect fastenings. Disconnect plugs and receptacles, multiple-line connectors in the electrical system, and detachable- and attachable-type fittings in hydraulic and pneumatic systems, as applicable, shall be utilized where practicable to permit component removal and replacement. All disconnect points shall be clearly indicated. Maximum practicable use shall be made of interchangeable hardware and fastening devices with a minimum number of types and sizes of bolts, screws, nuts, washers, and similar common (standard) parts. Lockwashers shall be secured to bolts or screws except where this practice is not practicable. Drain outlets shall be located for accessibility and shall permit the use of receptacles for collecting drainage. Drain plugs shall be removable by common wrench. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

37. Transportability.

3.7.1 Performance characteristics. Each unit shall meet the transport performance characteristics specified herein: (1) when prepared for transport; (2) when coupled to dolly and towed by a prime mover; and (3) when towed by a 5-ton, 6 by 6, M52 truck-tractor (see 3.27).

3.7.1.1 Highway operation. Each unit shall operate over smooth, hard-surfaced primary highways at speeds up to 20 mph.

3.7.1.2 Cross-country operation. Each unit shall operate over unimproved roads and open, rolling, and hilly cross-country terrain at speeds up to 10 mph.

3.7.1.3 Trailing and turning ability. Each unit shall trail the towing vehicle without weaving. Each unit shall follow the towing vehicle without exceeding 75 percent of the ICC towed-vehicle tolerance (see 3.5.1). Jack-knife turns up to an angle of 90 degrees for semitrailers and turns up to 60 degrees for dolly-semitrailer combinations as measured from straight-ahead to the right and to the left shall not cause damage to or interference between the towed unit, dolly, and the towing vehicle.

3.7.1.4 Braking ability. For each unit, the combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in conformance to ICC requirements, except that the stopping distance shall be accurately measured from initial movement of the brake pedal to stopping point. When actuated, the emergency breakaway system shall produce smooth emergency brake application.

3.7.1.5 Ground clearance. Exclusive of the running gear, the ground clearance of each unit shall be at least 12 inches.

3.7.1.6 Maximum height and width. The maximum allowable overall height and width of the semitrailers during highway operation (see 3.7.1.1) shall not exceed 126 inches and 102 inches, respectively.

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3.7.2 Railroad transport. Sectionalization for rail transportation shall be within the dimensional limits shown on figure 1. Sectionalization and reassembly of each unit shall each be accomplished in not more than 6 hours without cutting or welding.

3.7.3 Air transport. Sectionalization for air transport shall be within the following dimensional limits. Sectionalization and reassembly shall be accomplished as specified in 3.7.2.

Height - 102 inches.
Width - 102 inches.
Length - 32 feet 10 inches.

The angle of approach, interference, and departure shall be as defined in SAE J894, and when the dolly and each unit are coupled shall be as follows:

Angle of approach, minimum, front - 25 degrees.
Angle of interference maximum - 163 degrees.
Angle of departure, minimum, rear - 25 degrees.

3.7.4 Lifting and tiedown attachments. All sections separated to meet the requirements of 3.7.2 and 3.7.3 shall be provided with lifting and tiedown attachments conforming to 3.17 and 3.18, respectively.

3.8 Hoppers and chutes. All hoppers and chutes shall prevent spillage, and material shall flow steadily for continuous operation of the plant. All portions of hoppers and chutes subject to abrasive wear caused by movement of the aggregate shall be lined or faced with replaceable abrasion-resistant steel plates not less than 1/4 inch thick conforming to ASTM A242. All bolt heads subjected to the flow of material shall not extend beyond surfaces exposed to the material and shall be of hardened steel.

3.9 Diesel engines. Each diesel engine shall be of the naturally aspirated type. The engines, including all systems, components, accessories and auxiliaries shall conform to MIL-E-11276, except as otherwise specified herein.

3.9.1 Power and speed rating. The power and speed rating of each engine shall be such that operation of the plant under any of the operational requirements and conditions specified herein will not require horsepower in excess of the continuous horsepower rating of the engine for the applicable governed speed as established by the Qualified Products List, QPL-11276. The maximum engine torque at rated governed speed shall not exceed 90 percent of the peak engine torque available at a lower speed as determined by test method series 2000 of MIL-STD-1400 (see 4.6.1).

3.9.2 Fuel tank. A fuel tank conforming to MIL-E-11276 shall be furnished. The tank shall be furnished with a gasket, filler neck, and filler neck cap conforming to MS35643, MS35644, and MS35645. The tank shall have a minimum capacity sufficient for at least 10 hours operation under any operating condition specified herein. Fuel capacity shall be based on 85 percent of the fuel consumption rate obtained during conduct of tests in test method series 2000 of MIL-STD-1400 at applicable horsepower and governed speed.

3.9.3 Cranking system. A 24-volt cranking system and a 40-ampere battery charging system in accordance with MIL-E-11276 shall be furnished.

3.9.4 Instrumentation. An instrument panel with all instruments as specified in MIL-E-11276, except tachometer, shall be furnished.

3.9.5 Control systems. All control systems as specified in MIL-E-11276 shall be furnished. All

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controls shall be of the warning light type, except the overspeed control which shall be of the engine-shut-off type.

3.9.6 Cooling system. A complete cooling system in accordance with MIL-E-11276 shall be furnished. The radiator shall conform to MS35773.

3.9.7 Air cleaner. The air cleaner shall be of the dry type conforming to MIL-A-52363, class III. A rain cap shall be furnished. The air cleaner shall be provided with a restriction indicator conforming to MS53063-3.

3.9.8 Engine housing. Each engine shall be provided with a housing in accordance with MIL-E-11276. Weathertight seals shall be provided for operating doors and panels of the housing body. Material for the seals shall conform to ZZ-R-765, class Ia or Ib, grade optional, and shall be covered with nylon or other suitable material with a securing lip. Seals shall be riveted to the doors, panels, or housing after reinforcing with backup strip to prevent the rivet heads from pulling through the securing lip.

3.9.9 Clutch and clutch controls. The engine shall be equipped with an overcenter clutch which shall be accessible for lubrication, adjustment, and replacement of parts. The clutch shall withstand without deformation or damage repeated engagement under full load when tested as specified in 4.6.3.2. The lining shall conform to KKK-L-370, type II and shall be installed in accordance with the lining manufacturer's instructions. Clutch control shall be accessible from the operator's platform. Clutch capacity shall be not less than 150 percent of maximum engine torque.

3.10 Drives.

3.10.1 Chain drives. Unless otherwise specified herein, chains used in drive trains shall conform to MIL-C-52058, MIL-STD-421 or MIL-STD-423; or MIL-STD-424, as applicable. Chain drives shall be installed with idlers or other methods for providing adjustment.

3.10.2 V-belt drives. V-belt drives shall be in accordance with ASA B55.1, or Rubber Manufacturers Association, Inc., "Engineering Standard for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections)." V-Belt drives shall not exceed 8 feet center-to-center. Jack shaft drives may be used. All V-belt drives shall have provision for adjustment.

3.10.3 Gear drives. Gear drives shall conform to ASA B6.6, class 2 or class 3. Gear drives shall be totally enclosed and shall run in an oil bath. The enclosures shall be as specified in 3.15.4. Cast iron gears when used shall conform to ASTM A536. The analysis and the heat treatment shall be as recommended by the supplier. When cast iron gears are provided, the pinion shall be of steel.

3.10.4 Shafting. Shafting 2-inch diameter and larger shall be of steel in accordance with ASTM A193, type B7, Rockwell C hardness 25 to 32, and shall be turned, ground, and polished. The fit tolerance between all shafting and driven members shall not exceed 0.003 inch.

3.10.4.1 Keys and keyways. All keys and keyways, except for engines, speed reducers, clutches, and taper-lock bushings, shall be in accordance with figure 2; and key ends shall be rounded to prevent line or point contact after installation. Keys shall be at least as hard as the shafting.

3.11 Primary unit. The primary unit shall consist of a chassis upon which shall be mounted a reciprocating-plate feeder with loading hopper and grizzly bars, a bypass hopper, a 15- by 24-inch jaw crusher, a primary delivery conveyor for moving the material from the jaw crusher to the secondary unit, a diesel engine, drives, guards, operator's platform, and ladder.

3.11.1 Platforms and walkways. Platforms and walkways constructed of checkered plate, steel

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grating, or expanded metal shall be furnished to provide access to all points requiring routine maintenance and adjustment and for operational control as follows.

3.11.1.1 Operator's platform. An operator's platform not less than 3 feet wide and rectangular in shape shall be furnished at the forward end of the hopper and shall connect with walkways so as to provide a continuous walking area around the sides and forward end of the hopper. The walkways shall have a minimum clear width of 18 inches and shall extend from the operator's platform along both sides of the feed hopper. An access ladder to the operator's platform shall be provided as an integral part of the unit. Removable hand-rails shall be provided. The operator' platform and walkways shall be level.

3.11.1.2 Engine-servicing platform. A platform shall be provided from which the operator can service the engine. Provisions shall be made for holding and securing the platform for transport purposes.

3.11.2 Jaw crusher. The crusher shall be of the jaw type with overhead eccentric action. The opening at the top of the jaws shall be not less than 15 inches by 24 inches at nominal 3-inch jaw opening. The discharge opening at closed stroke shall be adjustable over a range from not more than 1-1/2 inches to not less than 4-1/2 inches with one toggle plate. The crusher shall be provided with a steel-plate hopper mounted over the crusher opening to receive material from the reciprocating-plate feeder and to charge the crusher. The jaw crusher shall crush materials ranging in size from a cross section of 13 by 22 inches down to the size of the minimum jaw setting.

3.11.2.1 Frame. The jaw crusher frame shall be constructed of cast steel or welded steel plate and shall withstand without damage or permanent deformation any stress caused by operation as specified herein. The frame, if of welded steel plate, shall be stress relieved after welding in accordance with the Standard Qualification Procedure of the American Welding Society for weldments of this size. The frame, if of cast steel, shall be in accordance with QQ-S-681, class 80-40 or higher. The contact points on the frame which support the parts that transmit the crushing stresses to the frame shall be machined, or provisions shall be made to distribute the crushing load on the frame so that the frame will withstand without damage or permanent deformation the stresses imposed when crushing maximum-size material.

3.11.2.2 Pitman. The pitman shall be made of cast or fabricated steel reinforced with ribs and machined to receive the movable jaw plate. If the pitman is constructed of cast steel, the steel shall be in accordance with QQ-S-681, class 90-60 or higher. If the pitman is of fabricated steel, the steel shall be stress relieved in accordance with the Standard Qualification Procedure of the American Welding Society for weldments of this size. At the lower, or toggle, end of the pitman, provisions shall be made to receive a renewable toggle seat.

3.11.2.3 Pitman shaft. The pitman shaft shall be an eccentric shaft machined from steel conforming to SAE 4140, 4340, or 6150, shall be heat-treated, and shall withstand the maximum crushing load without damage or permanent deformation. The main bearings shall be mounted in bearing housings on the crusher frame.

3.11.2.4 Pitman shaft bearings. The pitman and pitman shaft shall be mounted on antifriction bearings. The bearings shall be protected by labyrinth-type grease and dust seals which shall prevent dust and foreign matter from entering the bearings. The bearings shall conform to FF-B-185 or FF-B-187 except that taper bore bearings and taper sleeve adapters will be permitted.

3.11.2.5 Flywheels. Two gray-iron flywheels conforming to ASTM A48, class 35, 40, 45, 50, 55, or 60, shall be mounted on the pitman shaft and shall be counterbalanced to offset the mass of the eccentric revolving parts and be of sufficient size to provide inertia to transmit and equalize the energy necessary for crushing rocks having 13- by 22-inch cross section without varying the speed

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more than 10 percent. One flywheel shall be machine grooved for V-belts and used as a driving pulley. All flywheels shall be one-piece castings. The drive shall conform to 3.10.2.

3.11.2.6 Toggle plate. A cast iron or steel toggle plate shall be provided to rest between the toggle seat on the pitman and the seat in the toggle wedge of the adjusting mechanism. The toggle plate shall be held in place by means of a tension bolt and spring arranged so that the bolt will not bind or bend when in operation or during adjustment. The toggle plate shall be constructed so that it will break or deform when uncrushable material such as tramp iron enters the jaw crusher and shall fail before any other part of the jaw crusher is stressed enough to cause breakage or permanent deformation. The toggle plate shall not fail when crushing rock as specified herein.

3.11.2.7 Jaw and cheek plates. Jaw and cheek plates shall be made of manganese steel conforming to ASTM A128. Analysis and heat treatment shall be as recommended by the manufacturer of the jaw crusher. The moveable and stationary jaw plates shall be reversible and shall be secured so that they will not loosen and shift during the crushing operation. The back surfaces of the jaw plates shall be in a plane so that they will fit against their respective supporting member and provide equal load pressures in operation. The jaw plates shall be of the heavy-duty type and shall be not less than 2-1/2 inches thick at the main crushing area measured from the back bearing surface of the plate to the valley of the corrugations and shall be not less than 1-3/4 inches thick at the top and bottom of the plate measured in the same manner. The jaw plates shall be removable for replacement without the removal of other accessories or components except cheek plates and wedges. The jaw plates shall have inserts for attaching lifting eyes or similar means for removing or inserting new jaw plates. Cheek plates shall be mounted on the frame between both sides of the jaw plates to protect the frame during the crushing operation. The cheek plates shall be constructed of two or more pieces which shall be replaceable and shall not loosen and shift during the crushing operation.

3.11.2.8 Adjusting mechanism. An adjusting mechanism of the wedge-and-screw or shim type shall be provided which will permit adjustment of the opening between the jaws. The adjusting mechanism shall be cast steel or fabricated steel plate and shall be reinforced so as not to break or permanently deform when crushing maximum size material. All contact surfaces of the adjusting mechanism shall be machined to insure uniform distribution of the load on the respective parts of the jaw crusher frame. A stop shall be provided to prevent adjustment to actual contact of the jaws during the crushing cycle.

3.11.2.9 Jaw setting. The jaw setting shall be measured along or parallel to the longitudinal axis of the jaw crusher on a horizontal plane. Measurement shall be of the greatest opening in the minimum discharge position of the jaws in the crushing cycle, normally between the ridge of one jaw and the valley of the other. Variation between any two measurements for any jaw setting shall be not greater than 1/4 inch. Jaw settings shall be provided which can produce aggregate sizes specified in 3.11.2.

3.11.3 Reciprocating-plate feeder. A reciprocating-plate feeder of a size consistent with the size of the jaw crusher, with grizzly bars and a loading hopper, shall be provided. The feeder shall be mounted on the chassis and shall deliver material to the jaw crusher.

3.11.3.1 Loading hopper. The loading hopper shall be constructed of steel plate having a nominal thickness of not less than 0.250 inch. The hopper shall be not less than 5 feet 6 inches square at the top. The top of the hopper shall be not more than 10 feet 6 inches above the ground line when in operating position. The side plates of the hopper shall slope to a point which is not less than 4 and not more than 5 inches above the reciprocating plate of the feeder. The lower end of the hopper along three sides shall be vertical for sufficient distance to allow not more than 1/2-inch clearance between the hopper and the plate feeder. Replaceable wearing plates shall be attached to the two sides and to the end. The discharge end of the hopper shall be open and provided with a tie member

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across the end near the top to prevent the hopper from spreading. The tie member shall not interfere with charging of the hopper. The loading hopper shall be mounted over the reciprocating plate of the feeder. The hopper shall withstand without damage or permanent deformation normal power-shovel loading and shall also be adaptable to being fed with trucks or bulldozers from a ramp or quarry wall.

3.11.3.2 Feeder plate. The feeder plate shall be not less than 20 inches wide and 1/2 inch thick, reinforced with the structural members along two sides to form a pan with open ends. The feeder plate shall be supported on not less than two pairs of rollers or two shafts, each of which is provided with two integral, heat-treated, roller surfaces. The feeder plate with rollers, shafts, and bearings shall be supported on members connected to the structural legs. The reciprocating plate shall travel in a horizontal plane and shall have alternating forward and rearward motion activated by an adjustable eccentric driving mechanism providing a maximum stroke of 6 inches. The eccentric or crank mechanism shall be driven from the crusher shaft. The feeder drive shall be provided with an overload device and a hand-lever-operated friction clutch. The hand level shall be located within reach of the operator from the operator's platform.

3.11.3.3 Grizzly bars. Steel grizzly bars shall be provided. The bars shall be attached to the discharge end of the feeder plate and shall be spaced to accomplish separation and bypass of the undersize material while feeding the oversize material into the jaw crusher.

3.11.4 Bypass chute. A bypass chute shall be provided to receive the material passing through the grizzly bars and conduct it onto the receiving hopper of the primary delivery conveyor. The bypass chute shall be constructed in accordance with 3.8.

3.11.5 Primary delivery conveyor. The primary delivery conveyor shall be mounted under the jaw crusher and bypass chute so that functional clearance and inclination angle will permit delivery of the crushed and bypassed materials beyond the main chassis and into the hopper of the feeder conveyor on the secondary unit. The discharge end of the conveyor shall be hinged for transport purposes and a saddle with a tiedown device shall be provided to secure the free end of the conveyor when folded in the transport position. The conveyor shall be provided with skirtboards with adjustable flashing to prevent spillage of undersized material.

3.11.5.1 Frame. The conveyor frame shall be of rigid construction and shall support all other components of the conveyor not supported directly by the chassis. The frame shall support all dead and live loads imposed by other components when operating with a loading of 25 pounds per lineal foot of belt in the carrying run.

3.11.5.2 Conveyor drive pulley. The pulley shall be of steel construction not less than 26 inches long, not less than 8 inches in diameter, crowned, and mounted on antifriction bearings. The crown shall taper 1/8 inch per foot to the center of the pulley. The outside surface of the pulleys shall be concentric with the shaft within a total indicator reading of 0.060 inch, except that a total indicator reading of 0.090 inch is permissible in an area immediately adjacent to the longitudinal weld. The pulley supplying power to the belt shall be lagged.

3.11.5.3 Tail pulley. Unless used as the drive pulley, the tail pulley shall be of steel construction, crowned, wing, or self-cleaning type, and mounted on antifriction bearings. The pulley shall be not less than 8 inches in diameter and not less than 26 inches long. The outside surface of the pulley shall be concentric with the shaft within a total indicator reading of 0.100 inch. When the tail pulley is used as the drive pulley, it shall conform to 3.11.5.2.

3.11.5.4 Taper-type locking device (pulley and shaft). All pulleys shall be provided with an inner

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hub which clamps to the shaft by moving axially along a taper in the outer hub. The mechanical clamping shall be accomplished by the use of a screw member or members providing for mounting and removing the pulley.

3.11.5.5 Bearings. Bearings for pulleys, takeups, and drive shall be heavy-duty, antifriction type conforming to FF-B-171, FF-B-185, or FF-B-187. Bearings conforming to FF-B-171 may be retained on the shaft by a bearing-retainer plate and shims between the retainer plate and housing to provide endwise adjustment. When bearings conforming to FF-B-185 are mounted in a manner other than being secured against shoulders, the application shall be subject to the approval of the contracting officer for the specific application.

3.11.5.6 Takeup adjustment. To sustain the supplier's recommended belt tension, a takeup adjustment consisting of two 1-inch Acme thread arrangements (one per side) shall be provided at the tail pulley. The total available belt takeup on each side shall be not less than 12 inches. Threads shall conform to ASA B1.5.

3.11.5.7 Belt carriers. The conveyor belt shall be carried between pulleys on troughing rolls, impact rolls, and return rolls.

3.11.5.7.1 Troughing rolls. The troughing rolls and carrier assemblies shall conform to MS16961-1. The assemblies shall be mounted at intervals of not more than 4 feet.

3.11.5.7.2 Impact rolls. The impact rolls and carrier assemblies shall conform to MS16963-1. The assemblies shall be mounted at intervals of not more than 1-1/2 feet and shall support the belt under the receiving hopper, including the area under the jaw crusher.

3.11.5.7.3 Return rolls. The return roll assembly shall conform to MS16964-1. The assemblies shall be mounted at intervals of not more than 9 feet.

3.11.5.8 Conveyor belt. The conveyor belt shall be constructed in accordance with ZZ-B -206, type A, grade 375, 24 inches in width, with the following exceptions: The top cover shall be 1/8 inch thick, and the bottom cover shall be 1/16 inch thick with a skim coat between all plies. The friction pull shall be not less than 15 pounds. The tensile strength between top and bottom covers shall be not less than 2,000 psi. The belt fabric shall be treated to resist the growth of fungus.

3.11.5.8.1 Belt fasteners. The belt shall be furnished with bolted metal plate fasteners to provide a belt joint for assembly and disassembly of the belt.

3.11.5.9 Receiving hopper. The receiving hopper shall be mounted at the lower end of the conveyor frame and shall be removable. The hopper shall be of the skirtboard type and shall have three sloping sides to prevent spillage of material. The hopper shall extend not less than 12 inches beyond the discharge of the jaw crusher toward the discharge end of the conveyor. The space between the three sides of the hopper and the conveyor belt shall be provided with adjustable and replaceable rubber flashing mounted under the hopper plates. The flashing shall be 3/8 inch thick 5 percent wax impregnated, abrasion resistant, natural rubber compound of 2,500 to 3,000 psi tensile strength, 500 percent elongation, 40-50 durometer.

3.11.5.10 Drive mechanism. Power for the conveyor shall be supplied from the diesel engine specified in 3.9 by a drive as specified in 3.10 and may include a reducing gear. The drive shall be arranged so that the conveyor will run at all times when the jaw crusher flywheel is turning. A takeup, or takeups, shall be furnished for installation and for maintaining tension in V-belts or chains. If furnished, reducing gears shall be totally enclosed and shall operate in an oil bath. Oil temperature in bearings and gear cases shall not rise more than 100° F. above ambient temperature.

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3.11.5.11 Belt cleaner. A weighted or spring-loaded, plow-type, rubber-blade scraper shall be mounted inside the return run of the belt at the foot (tail) end of the conveyor. The scraper shall exert uniform pressure across the full width of the belt. The blades shall be neoprene rubber and shall be not less than 3/8 inch thick. The tensile strength of the blades shall be not less than 1,200 psi and the durometer hardness shall be 65 to 75.

3.11.6 Trailer.

3.11.6.1 Chassis frame. The framing members of the unit shall support the weights and maintain alinement under all conditions of transport and of plant operation as specified herein.

3.11.6.2 Towing pintle hook. A 40,000-pound towing pintle assembly conforming to MS51118 shall be mounted on the rear of the semitrailer in accordance with SAE J849, except that the brake and electrical connections shall be as specified in 3.11.6.7.1 and 3.11.6.9, respectively. The pintle shall provide a towing connection for the dolly converter when the unit is being towed by a truck-tractor.

3.11.6.3 Semitrailer upper fifth-wheel plate and kingpin. The upper fifth-wheel plate shall cover a 36-inch-diameter lower fifth wheel and the entire width of the gooseneck. The plate shall be fabricated of high tensile steel with not less than 50,000 psi yield point and shall have a thickness of not less than 0.370 inch. Both longitudinal and transverse reinforcing members shall be provided. At the front of the plate, a front coupling ramp shall be furnished. Ramps that are not flush with the front crossmember shall include at least four vertical gussets and shall be provided with holes for water drainage. An SAE 2-inch-throat-diameter kingpin conforming to MS53036 shall be installed not less than 24 inches or more than 36 inches from the front of the semitrailer. The center of the kingpin shall be within 1/8 inch from the longitudinal centerline of the semitrailer. The swing clearance rearward from the kingpin to any part of the semitrailer below the upper fifth-wheel plate shall be not less than 86 inches. The weight transferred to the lower fifth wheel shall not exceed 14,500 pounds on level ground.

3.11.6.4 Intervehicular connections. The semitrailer shall be furnished with airbrake couplings as specified in 3.11.6.7.1 and front and rear electrical intervehicular receptacles as specified in 3.11.6.9 for connecting the truck-tractor airbrake and electrical systems to the semitrailer systems, and the dolly airbrake and electrical systems to the semitrailer airbrake and electrical systems. The couplings and the receptacle shall be installed on the front of the semitrailer, symmetrical about the centerline in accordance with SAE J702. Clearance about the couplings and the receptacle shall permit connection of the truck-tractor hoses and harness to the semitrailer without interference with the upper fifth-wheel plate or other components. The airbrake couplings shall be protected with metal spring dust covers. The electrical receptacle shall have pigtail leads terminating within a splash-proof, semitrailer-type electrical nose box. The nose box shall also house the three circuit breakers as specified in 3.11.6.9 for the protection of the service-lighting circuits.

3.11.6.5 Stabilizing jacks. The semitrailer shall be equipped with not less than four screw-action stabilizing jacks, permanently attached to the chassis. The jacks shall conform to GGG-J-51, type I, class 3, diameter of the screw not less than 2 inches, and rated capacity 15-ton. The base of the jack shall be fully enclosed. The jack shall have a steel ball-bearing cap. Detachable pads may be furnished. Provisions shall be made for stowing the pads on the trailer in an identified, accessible location. The two front stabilizing jacks shall also serve as landing jacks and shall withstand without damage or permanent deformation the impact imposed when coupling the trailer with a truck-tractor.

3.11.6.6 Rear suspension. The unit shall be equipped with a single-wide-base-tire, tandem suspension system. The suspension system shall protect the crusher from shock and vibration and shall withstand static and dynamic loads without permanent deformation, misalinement or damage when the crushing plant is transported as specified herein. The suspension shall be of the

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elastomercushioned, walking-beam type having through axles; or the trunnion-padded, oscillating beam type having individual spindles. Wheel articulation shall provide not less than 8 inches difference in elevation between all combinations of both opposite wheels without loss of tire-to-ground contact. The suspension system shall include stops or bumpers of elastomer material to limit suspension movement at full compression and full rebound. The installation of the suspension shall be in alinement with the frame and with the axles or spindles so that the diagonal measurements from the true center of the kingpin to the center of each axle or spindle on the forward pair of wheels do not differ in excess of 1/16 inch. Each subsequent axle or spindle shall be parallel to the forward pair of wheels so that the suspension wheel bases on each side do not differ in excess of 1/8 inch. The maximum load imposed on the rear suspension, measured at the ground, shall not exceed 34,000 pounds. The rear suspension shall be equipped with complete assemblies of bearings, seals, hubs, and drums conforming to MS35865-C-1 and foundation brakes of the full-air type as specified in 3.11.6.7. Wheel location shall be matched to bearings. Suspension mounts including frame and through-axle types shall be installed in accordance with the suspension and axle manufacturer's recommendations. Tires, rims, valves, and metal valve caps shall conform to and be installed in accordance with the Tire and Rim Association, Inc., Yearbook. Wheels shall be size 13.00 by 22.5 and shall conform to SAE J876. The wheels shall be equipped with size 16.5 by 22.5, 16-ply rating, wide-base tubeless tires having traction-lug tread design. Tires shall include commercial antiozonant compounds. Tire and wheel assemblies shall be balanced within industry limits to eliminate wheel tramp and drumming at road speeds up to 45 mph. The unit shall be equipped with wheel splash fenders and guards which will restrict water and other road-surface substances thrown from the single-tire wheels. These components shall be in accordance with SAE J682, SAE J683, and MS51331.

3.11.6.7 Brake system. The semitrailer shall be equipped with a service and emergency-breakaway airbrake system complying with ICC Motor Carrier Safety Regulations except as specified herein. The brake system shall include:

(a) Service and emergency line filters.
(b) An SAE J10, end drain reservoir with 1-1/2 times the air volume recommended by SAE J813.
(c) A waterproof emergency relay valve conforming to MS53004.

All brake-system components shall be located, installed, and protected so that damage from ground obstructions and debris is prevented and so that maximum efficiency and ease of maintenance is obtained. The foundation brake assemblies shall be not less than 15- by 7-inch size.

3.11.6.7.1 Brake lines. The brake lines shall be routed within the protection of the semitrailer framing and shall be secured by coated tubing clamps spaced at intervals not to exceed 3 feet. Grommets and through-frame connectors shall be installed where lines pass through sheet metal and framing members. Brakeline components shall conform to the following:

(a) Flexible air hose	-	MIL-H-3992, type I.
(b) Rigid air line	-	SAE J844
(c) Fittings	-	Heavy-duty, airbrake, compression type, SAE J512.
(d) Intervehicular couplings	-	MS35746 and furnished with spring-loaded dust cover and identification tags conforming to MS53007.

3.11.6.7.2 Brake lining. The brake lining material shall contain a sintered, iron based lining mixture which shall be fused to a steel backing and installed on the brake shoes by the use of steel bolts or rivets as recommended by the lining manufacturer. The lining segment thickness, including the steel backing, shall be not less than 1/2 inch and shall be radial ground to fit the brake drum. The lining mixture shall conform to KKK-L-370, type II, except that the kinetic coefficient of friction at surface speeds of 4,000 to 6,000 feet per minute shall exceed 0.30, and other characteristics of the material shall be as recommended by the lining manufacturer for airbrake usage.

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3.11.6.7.3 Brake preservation. All metal surfaces enclosed by the brake drum including the interior surface of the drum, metal surfaces of the shoes, backing plates, and all attached parts shall be coated with a lacquer-resisting primer conforming to TT-P-664, or with a black oxide primer. Interior components of the brakes including metal surfaces of the shoes, backing plates, and all attached parts shall be primed with a black oxide primer. Wedges, pins, eccentrics, cam levers, anchors, and retracting springs shall be coated with preservative oil conforming to MIL-L-21260, grade 2. Items or surfaces which are wholly prime coated, painted in accordance with Government specifications, or constructed of materials providing corrosion protection equivalent to that provided by MIL-C-16173, grade 1, do not require additional protection.

3.11.6.8 Chock blocks and cradles. Four wood chock blocks with captive chains and four wire cradles shall be furnished. The cradles shall be installed, one forward and one to the rear on each side of the tandem axle assembly. The hook of each captive chain shall be snapped on the outer loop of each cradle to allow lifting the chock and the coiled chain from its cradle and placement of the chock under its respective dual tire. The chock blocks shall conform to figure 3 and the cradles shall conform to figure 4.

3.11.6.9 Vehicular electrical system. The unit shall be equipped with a waterproof, 24-volt, vehicular electrical system for service and blackout illumination during travel. The electrical system shall be in accordance with the SAE Handbook, the American Trucking Association (ATA) specification E-3, ICC Motor Carrier Safety Regulations, and Drawing 8347201, except as specified herein. All wiring and components shall be located and guarded to provide protection from falling and flying rock and other damage, breakage, and vibration. The electrical system shall be in accordance with table I. Item quantities that are not specified shall be furnished as required (AR) for each unit.

Table I. Vehicle Electrical System

Quantity	Description	Requirements
AR	Service light and reflector location.	ICC Regulations AND ATA SPEC. E-3
13 ea (min)	Service lights; clearance, side-marker and identification.	MS35423
8 ea	Blackout lights; clearance, side-marker.	MS35424
2 ea	Service combination lights.	SAE-STI, 4-in.-dia, waterproof.
2 ea	Blackout combination lights.	MS51330
8 ea	Reflectors.	MIL-R-12144
1 ea	Intervehicular harness assembly.	Dwg 7728811
1 ea	Front trailer receptacle.	MS75021-1
1 ea	Rear trailer receptacle.	MS75021-2
2 ea	Receptacle cover.	Dwg 7731428
AR	Receptacle installation.	Dwg 8722728
AR	Low-tension cable.	MIL-C-13486 and Dwg 7056674
AR	Waterproof cable insulation.	Dwg 8668076
AR	Harness insulation.	MIL-I-631 and MIL-I-15126
AR	Cable support straps.	MS21322, MS21333, and MS21334

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Table I. Vehicle Electrical System (Cont'd)

Quantity	Description	Requirements
AR	Circuit number charts.	Dwg 7070301
AR	Circuit number bands.	MS39020
AR	Friction retainment connectors.	MS27142 through MS27148 and MS33800
AR	Connector lubricant.	MIL-S-8660
AR	Connector holder clip.	Spring metal or plastic type.
AR	Threaded plugs and receptacles.	Dwg 7723494
AR	Terminal connectors.	Dwg 7056700
AR	Insulator sleeves.	Dwg 7056772
3 ea	Circuit breakers.	MS39062

3.11.6.9.1 Lighting. The semitrailer shall be equipped with both service and blackout lighting. The service lighting shall be as specified in table I and the blackout lighting shall be as specified herein using the wiring diagram as a guide. All lights shall be provided with shock and vibration protection by location and through use of resilient mountings to prolong filament life. Identification, side-marker, and clearance lights shall be clustered and mounted on plates inserted into lamp bracket sockets conforming to SAE J725. These clusters shall be removable for storage and identified with plates (see 3.21) to preclude incorrect replacement. Sidemarker and clearance lights, both service and blackout, shall be mounted on the same mounting at each corner. The rear service combination lights shall be of the flush mount, waterproof type having either a sealed unit mounted in an elastomer ring or a bulb and lens mounted in an elastomer housing. These lights shall be installed as far apart as practicable to indicate semitrailer width and for turn-signal operation. The blackout combination lights shall be installed toward the inside or below the service combination lights. These four lights shall be protected by recessing the lights within rings of 6-inch minimum diameter standard weight pipe.

3.11.6.9.2 Wiring. The wiring shall be fabricated and installed in accordance with the wiring diagram and with the "Electrical Equipment and Lighting" section of the SAE Handbook, except as specified herein. Cable clamps, grommets, loom, and commercial thin-wall conduit shall be used to prevent whipping, chafing, or other mechanical damage to the wires. Thin-wall conduit shall be provided where wires are not routed within the protection of framing members. Splices shall be limited to junctions within the harnesses and shall be waterproofed with insulation. Wires adjacent to friction connectors and junctions and two or more wires following the same routing and not protected by conduit shall be formed into a harness with either 0.010-inch-thick, 1-inch-wide tape or insulation. All wires shall be number coded near each connection following the circuit numbering charts and the wiring diagram and as specified herein. Friction connectors shall be lubricated with connector insulation prior to connection, and each assembly shall be secured to the semitrailer with connector holder clips. A splashproof, electrical, semitrailer-type nose-box shall be furnished to contain terminal connectors for all live circuits and the two ground circuits and for three service lighting circuit breakers. The circuit breakers shall be wired to protect circuits 22-460, 22-461, and 21-489. Each cluster light assembly shall be provided with two or three pigtail leads with friction connectors to provide live service and blackout or service only leads as applicable with ground return. The service combination lights shall be connected through pigtail leads with friction connectors to circuits 21, tail, and 22-460 or 22-461, stop and turn. The rear blackout combination lights shall be connected to circuits 24, tail, and 23, stop. The rear receptacle shall be wired into the system to provide lighting of other towed vehicles.

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3.11.6.9.3 Reflectors. Reflectors shall be furnished and mounted as required by ICC Motor Carrier Safety Regulations. In addition to meeting ICC regulations, two amber reflectors shall be mounted on the front of the semitrailer as far apart as practicable and at the same height as those mounted on the sides.

3.12 Secondary unit. The secondary unit shall consist of a chassis upon which shall be mounted a 24-inch-diameter by 16-inch-face roll crusher, a vibrator screen, a feed conveyor for moving the material from the delivery conveyor on the primary unit to the screen on the secondary unit, a return feed system for conveying crushed material to the screen for resizing, a fines delivery conveyor for conveying material into trucks or stockpiling, a delivery conveyor for conveying the finished material into trucks, a diesel engine, drives, guards, operator's platforms, walkways and a ladder.

3.12.1 Platforms and walkways. Platforms and walkways constructed of checkered plate, steel grating, or expanded metal shall be furnished to provide access to all points requiring routine maintenance, adjustment, or operational control.

3.12.1.1 Operator's platform. An operator's platform not less than 2 feet wide and extending along one side of the unit for accessibility and operation of the components shall be furnished. An access ladder to the operator's platform shall be provided as an integral part of the plant. Removable handrails shall be provided.

3.12.1.2 Engine-servicing platform. A platform shall be provided from which the operator may service and operate the engine. Provisions shall be made for holding and securing the platform for transport purposes.

3.12.2 Roll crusher. The roll crusher shall consist of two rolls with detachable one-piece roll shells, one fine-corrugated and one smooth, 24 inches in outside diameter, 2 inches in thickness, and with a face of 16 inches. The corrugated shell shall be mounted on the stationary drive roll. The shells shall be made of manganese steel conforming to ASTM A128 with analysis and heat treatment as recommended by the manufacturer of the roll crusher. Shells shall be mounted on heavy steel spiders or hubs by means of wedge-type, circular segments, or equal, so that the shell will not slip or shift and change the initial setting while in operation. Power for the roll crusher shall be supplied from the power-unit through V-belts to the main drive pulley. A guard shall be provided for the main drive. Adjustable compression springs acting against the sliding roll shaft housing or blocks with a capacity to provide pressure for crushing maximum size rock shall be provided. Provision shall be made to protect the driving mechanism from overload caused by entrance of foreign metallic objects between the rolls. The spring and roll adjustment shall be accessible for adjustment without dismantling other parts of the unit. A hopper shall be provided over the rolls to receive the materials delivered by the vibrator screen for secondary crushing. The hopper shall be provided with replaceable liner plates and shall be constructed in accordance with 3.8. The roll crusher frame shall be constructed of structural steel members braced, riveted, or welded to support all components of the crusher and shall withstand the stress of normal crusher operation without permanent deformation or damage.

3.12.2.1 Roll shaft bearings. The rolls shall be mounted on antifriction bearings. The bearings shall be protected by labyrinth-type grease and dust seals which shall prevent dust and foreign matter from entering the bearings. The bearings shall conform to FF-B-185, FF-B-187, or FF-B-171, except that taper bore bearings and taper sleeve adapters will be permitted.

3.12.2.2 Roll shaft. The shaft material shall be heat treated SAE steel 4140, 4340, or 6150.

3.12.2.3 Roll drive. The rolls shall be chain driven or driven and timed by star gears made of SAE heat-treated steel 4140, 4340, or 6150. The teeth on the star gears shall be of sufficient length so that the sliding-roll shaft may be adjusted over a total range of not less than 3 inches including a negative adjustment of 1 inch. The star gears shall be enclosed in a dustproof housing and shall run in oil. The

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stationary roll shaft shall be driven from the main drive shaft by means of a gear and pinion. The gear and pinion shall be cast or forged steel, with cut teeth, enclosed in a dustproof housing and shall run in oil. The face and pitch of teeth of the gear and pinion shall be of sufficient size to transmit the full horsepower of the engine when rock of maximum size is being crushed. A gray iron flywheel conforming to ASTM A48, class 35, 40, 45, 50, 55, or 60, shall be provided having sufficient inertia in operation to equalize sudden impact loads in crushing rocks having high compressive strength.

3.12.2.4 Stationary roll. The stationary roll shaft bearing assemblies shall be mounted on the frame or supporting frame or in the roll hub and shall be alined square with the roll frame.

3.12.2.5 Sliding roll. The shaft bearing housings of the sliding-roll assembly shall be mounted on the roll frame or in the roll hub and shall have a means for adjusting the opening between rolls.

3.12.2.6 Roll guard. A removable guard shall be provided. The guard shall be positioned over the rolls to prevent the hazard of flying rock.

3.12.2.7 Cheek plates. Cheek plates shall be mounted on the frame between both ends of the crusher rolls to protect the frame during crushing operations. The cheek plates shall be replaceable and shall not shift or loosen during the crushing operation. The cheek plates shall be made of manganese steel conforming to ASTM A128. The analysis and the heat treatment shall be as recommended by the manufacturer of the roll crusher.

3.12.3 Vibrator screen. The vibrator screen shall be either the inclined or the horizontal type, 3 feet wide by 8 feet long with two decks. The top deck shall divert oversize material to the roll crusher. The lower deck shall separate small-size material and fines. The top deck shall be furnished with 1-1/2-inch-square-opening screen cloth installed. The bottom deck shall be furnished with 1/4-inch-square-opening screen cloth installed. Screen cloth shall be as specified in 3.12.3.3. The vibrating screen shall consist of a stationary frame upon which shall be mounted a live frame with screen cloths, a vibrating mechanism, a screen feed hopper, and a screen discharge hopper and chutes.

3.12.3.1 Stationary frame. The stationary frame shall be constructed of structural members or castings and shall be securely mounted on the roll-crusher chassis so as to minimize the vibration transmitted to the roll crusher.

3.12.3.2 Live frame. The live frame of the screen assembly shall consist of steel plates on structural steel members constructed to hold the replaceable screen cloth (see 3.12.3.3) securely in place to prevent movement of the cloth during full-load operations. A series of longitudinal support bars shall be provided to create a crown of not less than 1/2 inch in the screen cloth. A molded-rubber cushion shall be provided between the screen cloth and the longitudinal supports. Provisions shall be made for changing screen cloth by using common tools. The live frame shall be mounted and balanced by means of springs, supporting rubbers, or the equivalent to provide an even flow of material across the full width of the cloth.

3.12.3.3 Screen cloth.

3.12.3.3.1 Material. All screen cloth shall be made of crimping-grade, oil-tempered wire conforming to National Sand and Gravel Association (NSGA) Circular No. 80, table 3, grade B, carbon range 0.55 to 0.70, drawn and hardened as specified.

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3.12.3.3.2 Wire size. Wire size shall conform to NSGA Circular No. 80, table 7, standard heavy, and shall be limited to the following:

Clear opening minimum (inches)	Wire diameter (inch)
2-1/2	1/2
1-1/2	3/8
3/4	1/4
1/4	0.135

Screen cloth shall be furnished in 3-foot by 8-foot, or 3-foot by 4-foot sections and shall conform to MS39127, except for length. One 8-foot section or two 4-foot sections of each of the above sizes shall be furnished in addition to those specified in 3.12.3.

3.12.3.4 Vibrating mechanism. A mechanical vibrating mechanism shall be provided for activating the live-frame assembly. When material from the deck or decks of the screen impinges on the activating mechanism, the mechanism shall be equipped with a replaceable, protective cover.

3.12.3.4.1 Bearings. All shaft bearings on the vibrating screen shall be the double-row, self-alining, antifriction type. All bearings exposed to dust shall be provided with labyrinth-type grease and dust seals to prevent leaking and to prevent dust entering the bearings.

3.12.3.5 Screen feed hopper. A hopper shall be mounted on or over the vibrator screen to receive material from the feed conveyor and to discharge it evenly across the full width of the screen.

3.12.3.6 Screen hopper and chutes. A screen hopper and two discharge chutes as specified in 3.8 shall be located under the vibrator screen to receive the material from the bottom-deck screen and to deliver it to a rock-delivery chute and a fines-delivery chute. The hopper shall have two compartments, one for coarse aggregate and one for fine aggregate. The compartments shall be separated by a manually operated adjustable flop gate or baffle. By adjusting the flop gate it shall be possible to direct all material to a single delivery chute.

3.12.4 Feed conveyor. The feed conveyor shall have a steel channel-type frame and shall be constructed in accordance with 3.11.5.1 through 3.11.5.11. The conveyor shall be mounted with the discharge end extending over the vibrator screen and shall receive material from the primary delivery conveyor of the primary unit and crushed material from the return-feed system and shall deliver these materials to the vibrating screen for resizing. The conveyor shall be mechanically driven from the head-end through a countershaft on the plant. The drive shall be in accordance with 3.10.

3.12.5 Return-feed system. The return-feed system shall deliver the crushed material to the feed conveyor as specified in 3.12.4 and onto the vibrator screen for resizing. The return feed system shall be of the rotary elevator type, or of the side-return conveyor type with a cross-feed conveyor. If the side-return conveyor type is used, the side conveyor and the cross-feed conveyor shall be constructed in accordance with 3.11.5.1 through 3.11.5.11, except the cross-feed conveyor may be of the flat-belt type with straight idler rolls in lieu of troughing rolls.

3.12.5.1 Rotary elevator. The rotary elevator, if used, shall be friction or mechanically driven by the engine through a countershaft. The mechanical drive shall be as specified in 3.10. The rotary shaft shall be mounted in antifriction bearings supported on a steel frame on the chassis. Bearings shall be as specified in 3.11.5.5. The bucket flights shall be replaceable and fabricated from abrasion-resistant steel conforming to ASTM A242. A locking device shall be provided to prevent movement while the unit is being transported.

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3.12.6 Rock-delivery conveyor. The rock-delivery conveyor shall be constructed in accordance with 3.11.5.1 through 3.11.5.11, except the frame shall be in hinged sections to permit being folded up against the side of the unit by hand winch or other suitable means. Sling chains or bolts shall be provided to hold the conveyor securely in the folded position against the side of the unit for transporting. The conveyor shall be driven through a friction clutch and by mechanical drive from a countershaft. Mechanical drive shall be as specified in 3.10. The clutch shall be controlled from the operator's platform. The conveyor, when in operating position, shall be located with the lower end under the finished product side of the screen hopper and extended out one side of the unit a sufficient distance to provide a discharge height from ground level to centerline of head pulley of not less than 9 feet.

3.12.7 Fines delivery conveyor. The fines delivery conveyor shall conform to 3.12.6 and shall be mounted on the opposite side of the unit from the rock-delivery conveyor.

3.12.8 Trailer. The semitrailer shall be as specified in 3.11.6.

3.13 Front dolly converter. A dolly converter shall be furnished with each semitrailer-mounted unit. The dolly design shall permit coupling the dolly to the semitrailer by backing and shall permit cross-country travel.

3.13.1 Fifth wheel. A heavy-duty, 33-inch-disk, lower fifth wheel shall be furnished. The configuration of the fifth wheel and its mechanism shall be in accordance with MS53033. The fifth wheel and its mechanism shall accept and couple with the 2-inch throat diameter kingpin specified in 3.11.6.3. Ramps shall be provided to guide the kingpin into the fifth wheel when backing. The height of the loaded fifth wheel shall be 56 inches plus or minus 2 inches. The mounting of the fifth wheel shall be in accordance with ICC regulations.

3.13.2 Lunette and towbar frame. A lunette conforming to MS51337 shall be installed in a housing integral with the towbar frame. The lunette shall be supported the entire length of the shank. The clearance between the shank and support shall not exceed 0.035 inch. When the dolly is in towing position, the height of the lunette shall be between 34 and 36 inches from the ground line. The center of the lunette and the centerline of the towbar shall be located on the longitudinal centerline of the dolly within 1/8 inch. The dolly towbar shall be integral with the frame. When coupled to the semitrailer, the dolly shall turn 110 degrees both to the right and to the left of the longitudinal centerline of the semitrailer. The lunette and towbar assembly shall be compatible for towing with a military, 5-ton, 6x6, M51 dump truck as shown on MS500010; or a commercial dump truck having not less than 32,000 pounds GVW rating and full trailer towing components in accordance with SAE J849, as specified (see 6.2), and shall permit the centerline of the truck to be at any angle up to 60 degrees to the right and to the left of the centerline of the dolly when in towing position. There shall be no interference between the truck and the dolly-semitrailer combination.

3.13.3 Landing leg. The towbar of the dolly converter shall be provided with an adjustable, folding, landing leg to permit coupling and uncoupling of the lunette and the towing pintle by one man. Ground clearance shall be not less than 18 inches when the towbar and jack are in travel position.

3.13.4 Dolly suspension. The dolly shall be supported on a sprung single-axle assembly having wide-base single tires. The suspension components and the installation, when coupled between a truck and the unit, shall withstand towing and braking forces, cross-country articulation, and shock loads under the conditions specified herein without permanent deformation or damage. The suspension and fifth-wheel oscillation shall provide not less than 8 inches articulation between tires of the dolly as measured from the level unit. The axle assembly shall be of the through-axle type and as specified in 3.11.6.6 and 3.11.6.7 through 3.11.6.7.3, except that the brake chambers on the dolly

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axle shall be at least one size smaller than the chambers on the rear axles or a reducing valve shall be furnished to provide greater braking effort on the rear brakes than on the dolly brakes to reduce jack-knifing tendencies.

3.13.5 Dolly electrical system. The electrical system of the dolly shall conform to 3.11.6.9 using Drawing 8347066 as a guide in lieu of Drawing 8347201. Service and blackout rear combination lights shall be furnished and installed.

3.13.6 Dolly intervehicular components. The dolly shall be provided with electrical and airbrake lines to connect the truck systems to the semitrailer systems. All intervehicular components between the dolly and the truck shall extend 36 to 42 inches toward the truck from the center of the lunette eye. All intervehicular components between the dolly and the semitrailer shall be supported on a truck-tractor-type, flexible, spring pole. The pole shall be spring loaded to permit repeated bending to the horizontal position without damage. The spring pole shall prevent entangling or damaging the lines by either the dolly or the unit. The intervehicular component shall have a minimum length of not less than 10 feet from the dolly frame to the connections and shall allow the dolly to turn at least 110 degrees to the right and 110 degrees to the left from straight ahead. Clips, dummy couplings, and eye bolts shall be mounted on the dolly to protect and to hold the intervehicular components during storage, operation, and travel when the component is not in use. Electrical components shall be as specified in 3.11.6.9. Airbrake components shall conform to 3.11.6.7 and shall include hose coil spring guards on the couplings.

3.13.7 Safety chains. Two safety chains fitted with grab hooks shall be furnished. The chains shall extend between 36 and 42 inches beyond the center of the lunette eye. The chains shall be in accordance with ICC Motor Carrier Safety Regulations, except that the safety-chain hooks shall fit the shackles of the towing vehicles and of the semitrailer (see 3.11.6.2).

3.14 Electromagnetic compatibility. The electromagnetic compatibility of the plant shall be in accordance with MIL-E-55301, tactical support class.

3.15 Lubrication. All surfaces requiring lubrication shall be provided with a suitable means for lubricating.

3.15.1 Lubricants. The plant shall operate as specified herein when lubricated with military lubricants (see 6.3).

3.15.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where oil holes are applicable, they shall have spring-loaded covers. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.15.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.15.4 Enclosure with integral reservoir of lubricant. Enclosures such as gearcases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, finger holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped with a magnetic drain plug not less than 1/2-inch pipe size conforming to MS35844 or MS49006 and with a means for filling the enclosure with lubricant. The drain plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the plant is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the

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ground. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts except that panels and plates equipped with hand-operable quick-disconnect fasteners may be removed.

3.15.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The plant shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.15.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0°F. to 32°F. Each lubricated component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch to component or element surfaces which have been previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.17 Lifting attachments. Each unit and each sectionalized component (see 3.7.2 and 3.7.3) shall be provided with permanently affixed lifting attachments. The attachments shall enable the unit or component to be lifted in its normal traveling or operating position. The attachments shall be located so that not less than a 1-inch clearance is maintained between the slings and all exterior parts of the item being lifted. Spreader bars are permitted. Each attachment shall withstand without damage to any part of the plant not less than 2-1/2 times the normal load imposed on the attachment. Attachments for multiple slings shall be located so that the slings will converge not more than 19 feet above the lowest extremity of the load. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. The inside diameter of the eye shall be not less than 3 inches. Members to which attachments are fastened and the fasteners shall withstand without damage or permanent deformation all stresses imposed by the attachment. Where required, structural members shall be reinforced to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.18.

3.18 Tiedown attachments. Each unit and each sectionalized component (see 3.7.2 and 3.7.3) shall be provided with integral attachments to permit tiedown to the floor or deck of the transportation medium. Each attachment shall withstand, without damage to the plant, its proportion of the following static loads: 8-G forward, 8-G rearward, 2-1/4-G upward, and 1-1/2-G sideward, where G equals cargo weight of the plant. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.17.

3.19 Identification marking. The plant shall be identified in accordance with MIL-STD-130.

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Identification plates shall conform to MIL-P-514, type I, style 2, photographic plate. Transportation-data plates conforming to MIL-P-514, type V, photographic plate, shall also be furnished and shall indicate the location and capacity of lifting and tiedown attachments.

3.20 Treatment and painting. The portions of the plant and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.21 Instruction plates. Each unit shall be equipped with instruction plates or diagrams, including warnings or cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the plant. The words "WARNING" and "CAUTION" shall be red, number 31136, conforming to FED. STD. No. 595. Instruction plates shall conform to MIL-P-514, type IX, photographic plate.

3.22 Vehicle marking. The marking of each semitrailer and dolly shall be in accordance with MIL-STD-642. The gross vehicle weight of each semitrailer and dolly and the weight of each sectionalized component shall be marked on the side in such a manner as to be discernible. The prescribed tire pressure shall be marked near the tires. Lifting and tiedown attachments shall be marked "LIFT HERE" or "TIEDOWN HERE" as applicable.

3.23 Technical Publications. Such technical publications as are specified shall be furnished (see 6.2).

3.24 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3.25 Toolbox. Each unit shall be provided with a weatherproof toolbox made of steel not less than 0.071 inch thick. The toolbox lid shall be mounted on hinges conforming to MS35829 with a 0.125-inch-diameter brass pin. The toolbox lid shall open to at least 90 degrees from the closed position. The lid shall remain in the fully open position either by its own weight or by holding devices. Two hook-type fasteners conforming to Drawing 7969299 shall be provided to keep the lid closed. Not less than a 3/4-inch clearance shall be provided between the hook lever and any other part of the toolbox. The fasteners shall be attached to the toolbox by either bolts or rivets. A hasp which will accommodate a lock conforming to MS35647 shall be provided for locking the toolbox. The toolbox shall be fastened to the trailer or semitrailer in a protected position. A drain hole with rubber plug shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the tools specified (see 3.24).

3.26 Items to be furnished with the plant. No shipment of plants shall be made unless repair parts, tools, and accessories are included with the shipment or unless approval for shipment of the plant without such items has been received from the contracting officer.

3.27 Government-loaned property. The following property in the quantities indicated will be loaned by the Government (see 6.4):

Item No.	Description	Identification	Quantity
1	M52 Truck-Tractor, 5-ton 6x6	Mil-T-740	1
2	M51 Truck, Dump, 5-ton 6x6	MIL-T-740	1

Note: Item 2 will be loaned only when compatibility with the M51 dump truck is specified (see 3.13.2).

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3.28 Workmanship. The plant shall be free from defects such as incomplete welds, rust, cracks and other defects that could impair the operation of the plant. Metal used in the fabrication of the plant shall be free of kinks and sharp bends. Shearing and chipping shall be done neatly and accurately. Flame cutting of steel may be employed. Burned surfaces of flame-cut steel shall be ground or machined to remove all ash, burrs, and rough edges. All bends shall be made with metal dies or fixtures to insure uniformity in size and shape. All threaded parts shall be in accordance with Handbook H28 unless otherwise specified herein. Rivets shall fill the holes completely. Rivet heads shall be full, neatly made, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.28.1 Welders and welding.

3.28.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

3.28.1.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are tested as specified herein.

3.28.2 Surfaces. All parts, components, and assemblies including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be cleaned and free of sand, dirt, fins, pits, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled.

3.28.3 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings. Unless prior approval is obtained from the contracting officer, all tapped holes shall be countersunk before tapping.

3.28.3.1 Line boring for main bearings on jaw and roll crusher. Bearing support surfaces for each shaft shall be machined in a single setting to a tolerance of not more than 0.001 inch total indicator reading per foot of shaft. External bearing supports shall be doweled or provided with an eccentric arrangement to assure reassembly in its initial position prior to final machining and installation of the bearings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The plant shall be examined as specified in 4.6.2. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The engines to be utilized in the preproduction model and the plant shall be tested as specified in 4.6.1 and 4.6.3 through 4.6.3.6. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. An initial production plant(s) will be selected at random by the Government from the first five plants being produced from production tooling and will be examined as specified in 4.6.2 and tested as specified in 4.6.3 through 4.6.3.5 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production plant shall not exclude the remaining plants from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of an initial production plant to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production plant(s) and shall be cause for refusal by the Government to continue acceptance of plants until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on plants previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all plants accepted prior to the completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each plant shall be examined as specified in 4.6.2. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each plant shall be tested as specified in 4.6.3.7. Failure of the test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Engine performance. Prior to installation of the engines in the preproduction model, each engine shall be tested in accordance with MIL-STD-1400, test method series 2000. Nonconformance to any test requirement shall constitute failure of this test.

4.6.2 Examination. The plant shall be examined for the following defects (items 114 through 118 shall be examined while the plant is operating):

101. Material not as specified.

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102. Parts or components missing or do not function.
103. Assembly is incomplete or incorrect.
104. Safety devices or guards missing.
105. Workmanship not as specified.
106. Ease of maintenance not as specified.
107. Dimensions or weight not as specified.
108. Treatment and painting not as specified.
109. Identification or other marking, operating, and safety instruction, or manuals missing.
110. Lubrication devices not accessible.
111. Brake preservation not as specified.
112. Brake system not as specified or does not function.
113. Electrical system not as specified, does not function, or is damaged.
114. Oil or grease seals leaking.
115. Bearings or gear boxes overheating.
116. Shaft misalinement.
117. Belt carrier rolls do not turn.
118. Belt does not run straight on the centerline of the conveyor.

4.6.3 Tests.

4.6.3.1 Test conditions (operating). Prior to test, the plant shall be lubricated with military oils and greases as specified in 3.15.5 and 3.15.6. Oils shall be those designated for use in the ambient temperature at which the test is being performed. Quarry rock shall be as specified in 3.1.1.1.

4.6.3.1.1 Percentage of crush. A representative sample (not less than 50 tons) of the material being fed to the primary unit shall be taken and the percent of material larger than 1-1/2 inches shall be determined. Feed containing more than 20 percent of material smaller than 1-1/2 inches shall not be used.

4.6.3.1.2 Operating time. Crushed rock production capacity shall be computed on the basis of net operating time which shall be defined as gross operating time less allowable downtime for the following:

(a) Jamming of jaw crusher from rock fragments larger than the maximum size of rock specified in 3.11.2.
(b) Normal machinery maintenance. (Downtime due to any manufacturing deficiency shall not be considered normal maintenance.)
(c) Delay due to lack of material delivery to the plant.
(d) Other conditions considered by the contracting officer to be beyond the control of the supplier.

4.6.3.2 Capacity and performance. Operate the plant under the test conditions specified in 4.6.3.1 through 4.6.3.1.2 for 100 hours. Inability of the plant to operate under the specified test conditions or as specified in 3.1.1 shall constitute failure of this test.

4.6.3.2.1 Flywheel. Measure the speed of the jaw crusher flywheel during the capacity test for a continuous period of 3 minutes. Select a 3-minute period during which at least one rock having 13-by 22-inch cross section is being crushed. Nonconformance to 3.11.2.5 shall constitute failure of this test.

4.6.3.2.2 Uncrushable material. With the plant operating at the capacity specified in 3.1.1, perform the following:

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(a) Drop a piece of uncrushable material such as tramp iron into the jaw crusher. The piece shall be larger but not more than 2 inches larger than the maximum jaw opening at the jaw setting being used.
(b) With the roll crusher set to produce 1-1/2-inch aggregate, pass a piece of uncrushable material having a minimum dimension of not less than 3 inches through the rolls.

Nonconformance to 3.11.2.6 or 3.12.2 shall constitute failure of this test.

4.6.3.3 Transporting. All transporting tests shall be conducted on each unit of the plant.

4.6.3.3.1 Highway and cross-country. Using a Government-loaned M52 truck-tractor (see 3.27), tow the unit (in normal transport condition) over highways, or equivalent roadway, for a distance of 250 miles at speeds up to 20 mph and over cross-country terrain as specified in 3.7.1.2 for a distance of 50 miles at speeds up to 10 mph. Using the dolly converter specified in 3.13 and the dump truck specified in 3.13.2 instead of the M52 truck-tractor, repeat the highway and cross-country tests. Before any highway and cross-country tests are conducted, measure all critical mounting points on each unit. After all highway and cross-country tests are completed, check the measurement of critical mounting points and magnaflux all welds. Damage to or permanent deformation of any part of the plant, change in critical mounting point dimensions, evidence of any weld being defective, incompatibility with any towing vehicle, or inability to complete any portion of the test shall constitute failure of this test.

4.6.3.3.2 Truck and dolly-semitrailer compatibility. With the dump truck specified in 3.13.2 and dolly-semitrailer in towing position, jacknife the truck-dolly-semitrailer combination until the centerline of the truck is at an angle of not less than 60 degrees with the centerline of the dolly. Next tow the dolly-semitrailer combination at minimum truck turning radius to the right and to the left. Test all lights. Repeat the tests using the Government-loaned M52 truck-tractor in lieu of the dump truck. Interference between either truck and any part of the dolly or semitrailer or any light not functioning shall constitute failure of this test.

4.6.3.3.3 Truck-tractor and semitrailer compatibility. Couple the Government-loaned M52 truck-tractor to the semitrailer and tow the combination at the minimum turning radius of the truck to the right and to the left. Incompatibility with the truck-tractor, or interference with the truck-tractor by any part of the semitrailer shall consitute failure of this test.

4.6.3.3.4 Dolly and semitrailer compatibility. With the dolly and semitrailer in travel position, hold the lunette between 34 and 36 inches from the ground and turn the towbar 110 degrees to the right and to the left of the centerline of the semitrailer. Interference between the dolly and any part of the semitrailer shall constitute failure of this test.

4.6.3.3.5 Railroad and air transport. Sectionalize and reassemble each unit. Nonconformance to 3.7.2, 3.7.3, or 3.7.4 shall constitute failure of these tests. The tests for sectionalization and for lifting and tiedown attachments may be accomplished in conjunction with the tests specified in 4.6.3.4 and 4.6.3.5.

4.6.3.4 Lifting and tiedown attachments. Apply the static load specified in 3.17 for lifting attachments and 3.18 for tiedown attachments to the attachments after the attachments are mounted on the unit or section, as applicable. The direction of the applied load shall be in the direction in which the load is normally applied. Weld failure or permanent deformation of the attachment or of any component of the unit or section shall constitute failure of this test.

4.6.3.5 Tiedown attachments (dynamic). Anchor each section of each unit independently, as sectionalized for rail transport, to a flat car using only the tiedown attachments. Use no blocking. Hump the flatcar at a speed of 7 miles per hour against five box cars with their brakes set by air.

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Repeat the test with the opposite end of the section in the direction of the five cars. Any permanent deformation or breakage of the tiedown attachments or parts of the section shall constitute failure of this test.

4.6.3.6 Electromagnetic compatibility. The preproduction model, equipped for the reduction of radio interference in accordance with 3.14 shall be tested as specified in MIL-E-55301. The supplier shall furnish the contracting officer the reports required by MIL-E-55301, except that the time interval for submission of reports shall be as specified (see 6.2). Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production plants.

4.6.3.7 Production test. Operate the plant for a period of 1 hour without crushing rock. All adjustments shall be made which are required to begin production of stone. Jaw and roll settings shall be to produce a 1-1/2-inch nominal product. Any operational difficulties, breakage, or permanent deformation of any component of the plant shall constitute failure of this test.

4.7 Inspection comparison. The Government may select plants at any time during the contract production period and subject these plants to the examination specified in 4.6.2 and the tests specified in 4.6.3 through 4.6.3.4 to determine that the quality of the selected plant is equal to the quality standards established during initial production testing. The inspection will be performed by the Government at a site selected by the Government. Plants will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production plant(s).

4.7.1 Inspection failure. Failure of any plant when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of plants until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. The preproduction pack shall be examined as specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Tests. The preproduction pack shall be tested in accordance with 4.8.1.2.1 and 4.8.1.2.2. Failure of the railcar impact test or the pendulum impact test shall be cause for rejection.

4.8.1.2.1 Railcar-impact test. The crate shall be securely blocked to prevent movement on the car. Antiskid plates shall not be used for this test. The car shall be humped or shunted to cause impact at not less than 10 miles per hour striking a string of five empty cars with brakes set and draft gear extended. A single impact shall be applied to each side and each end of the crate (a total of four impacts).

4.8.1.2.2 Pendulum-impact test. The crate shall be swung as a pendulum against a flat, rigid wood, concrete or masonry wall or equally unyielding flat vertical barrier that is oriented perpendicular to the line of swing. The crate shall be suspended in a position 1 to 3 inches above the floor by not less than four ropes, chains, or cables whose upper ends are fastened not less than 16 feet above the floor. In the rest position, the end of the crate shall just touch the barrier. The crate shall be pulled back from the rest position, perpendicular to the face of the barrier until the center of gravity is raised vertically 9 inches. The crate shall be released from this position and permitted to swing freely against the barrier. A single impact shall be applied to each side and each end of the crate (a total of four impacts).

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4.8.1.2.3 Criteria for failure. Any shifting of contents, loose or broken holddowns, ties, stays, blocking, or bracing, or any visible damage to the crated components contributable to the preproduction pack shall constitute failure of this test.

4.8.2 Quality conformance inspection.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

119.	Materials not as specified for level A. Each incorrect material shall constitute one defect.
120.	Sectionalization not as required for level A.
121.	Disassembly not as specified for level A.
122.	Parts and sections not matchmarked for level A.
123.	Cloth tags and printing not waterproofed for level A.
124.	Unpainted exterior metal surfaces not coated with preservative.
125.	Trailer chassis and dolly not preserved in accordance with the referenced standard for level A.
126.	Engine not preserved in accordance with the referenced specification for level A.
127.	Preservative contacts the clutch lining for level A.
128.	Belts not removed or released from tension for level A.
129.	Strapping for intermediate containers not zinc coated for level A.
130.	Components secured to the unit interfere with lifting or mobility of the unit for level A.
131.	Blocking and bracing for level A not as specified.
132.	Strapping not zinc coated for level A packing.
133.	Marking illegible, incorrect, or incomplete for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The preproduction pack is a pack which will be identical to the pack which the supplier will subsequently produce in fulfillment of the contract. A preproduction pack shall be made and examined as specified for the applicable level to determine conformance to the preservation, packaging, packing, and marking requirements. Approval of the preproduction pack shall not relieve the supplier of his obligation to preserve, package, pack, and mark the plants in accordance with this specification. Any changes or deviations of production packs from the approved preproduction pack shall be subject to the approval of the contracting officer.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Sectionalization and disassembly. Each unit shall be sectionalized to meet the requirements specified in 3.7.2 and 3.7.3 as specified (see 6.2). In addition, components of units and sections vulnerable to damage, pilferage, and loss and protruding parts that increase cubage and that can be removed and reassembled without special tools shall be removed. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss.

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5.2.1.2 Matchmarking. To facilitate reassembly, components and sections removed from the units shall be matchmarked. Large sections and components shall be matchmarked identically by stenciled letters or numerals with lusterless white enamel overcoated with varnish conforming to TT-V-121, or identically matchmarked with paint conforming to MIL-P-13983. Removed small components shall have matchmarking information on cloth tags conforming to UU-T-81, type A, or on metal tags. The cloth tags and printing shall be waterproofed in accordance with MIL-STD-129. The tags shall be securely attached to the components so as to prevent damage to the preservation or to the components.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unpainted surfaces. Unpainted exterior metal surfaces of components and parts of the plant such as hoppers, chutes, basket flights, grizzly bars, feeder mechanism, conveyors, rolls, screens, shafts, rods, clevises, nonprecision chains, pulleys, ratchets, pins, linkages, drawbar, levers, bolts and nuts, and similar parts not specified herein requiring a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. Unpainted metal identification and instruction plates shall be coated with varnish conforming to MIL-V-13811. Machined surfaces shall be coated with type P-6 preservative and wrapped or covered with barrier-material conforming to MIL-B-121, type I, grade A, class 2, and the wrap or cover shall be secured with tape conforming to PPP-T-60, class 1.

5.2.1.5 Trailer chassis and dolly. The components of the semitrailer chassis and dolly shall be preserved and packaged in accordance with MIL-STD-281, level A (mobile).

5.2.1.6 Engine. The diesel engines and engine accessories shall be preserved in accordance with MIL-E-10062, level A, method I.

5.2.1.7 Clutches.

5.2.1.7.1 Disk-type. The inspection plates shall be removed. With the clutch engaged, all interior components of the clutch shall be sprayed with primer conforming to TT-P-664. Coating of nonmetallic surfaces shall be kept to a minimum. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutches shall be blocked in a partially disengaged position to eliminate contact between the disk facings and pressure plates. Snap-over-center and toggle-in-type clutches shall be completely disengaged.

5.2.1.7.2 Band and shoe type. The clutch drum facings shall be coated with a thin film of primer conforming to TT-P-664. Clutch-control mechanisms enclosed within housings shall be coated with type P-10, grade 1 preservative. Clutch control mechanisms not enclosed within housings shall be coated with type P-1 preservative. Preservative shall not contact the clutch lining.

5.2.1.7.3 Cone-type clutches. The unlined cone and clutch-control mechanisms shall be coated with a thin film of primer conforming to TT-P-664. The clutch shall be disengaged to prevent the lining from adhering to the unlined cones.

5.2.1.8 Gears.

5.2.1.8.1 Enclosed gears and transmission. When the gear lubricant contained in the gear housing conforms to the applicable lubrication order or to MIL-L-2105 or MIL-L-21260, the quantity required to bring the lubricant to the operating level shall be added. When the lubricant does not conform to any of the lubricants specified herein, the gear housing shall be drained and refilled to the operating level with the applicable lubricant. The gears shall be actuated to assure that all interior parts and surfaces are coated.

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5.2.1.8.2 Exposed gears. Exposed gears shall be coated with type P-1 preservative.

5.2.1.9 Chains.

5.2.1.9.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 preservative to insure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains shall be coated with type P-1 preservative. Unpainted surfaces of the sprockets shall be coated with type P-1 preservative. Sprockets shall be rotated during application of the preservative to insure complete coverage. Drive chains removed from the plant shall be preserved as specified herein and coiled, and the coil shall be wrapped with barrier-material conforming to MIL-B-121, type I, grade A, class 2, overwrapped with paper conforming to MIL-P-130, type I, and secured with tape conforming to PPP-T-60, class 1.

5.2.1.9.2 Enclosed chains. Enclosed chains shall be preserved as specified in 5.2.1.8.1 for enclosed gears.

5.2.1.10 Conveyors. Unpainted metal surfaces of rolls, shafts, belt tension screws, idlers, frame members, and hinges shall be coated with type P-1 preservative.

5.2.1.11 Conveyor belt fasteners. The belt fasteners shall be coated with varnish conforming to MIL-V-13811.

5.2.1.12 Belts and pulleys. Belts shall be removed or released from tension. Removed small belts shall be preserved in accordance with MIL-P-116, method IC-1. Removed large flat belts shall be compactly rolled and secured and wrapped with paper conforming to UU-P-271, class C-1, with all folds, seams and laps sealed with adhesive conforming to MMM-A-260, grade B, or with tape conforming to PPP-T-60, class 1. Unpainted pulley faces shall be coated with primer conforming to TT-P-664.

5.2.1.13 Keyways and keys. When pulleys or flywheels are removed from the plant, the keyways and keys shall be coated with type P-1 preservative and the key shall be secured in the keyway of the pulley or flywheel with tape conforming to PPP-T-60, class 1.

5.2.1.14 Technical publications. Technical publications required for each plant shall be preserved together in accordance with MIL-P-116, method IC-1. The preserved publications shall be placed in the toolbox.

5.2.1.15 Repair parts. Repair parts shall be preserved and packaged in accordance with MIL-P-116 and MIL-E-16298, as applicable.

5.2.1.16 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with MIL-H-15424, level A. The preserved and packaged tools shall be placed in the toolbox.

5.2.1.17 Intermediate packaging.

5.2.1.17.1 Components. The packaged and unpackaged components and parts that cannot be secured to the plant or to the sections shall be intermediate packaged in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional. Cushioning and blocking shall be provided to prevent movement and damage to the contents. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.2.1.17.2 Repair parts. Repair parts required for each plant shall be intermediate packaged in boxes as specified in 5.2.1.17.1.

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5.2.2 Level C. The plant components, repair parts, tools, and publications shall be preserved and packaged to afford protection against deterioration and damage from the supply source to the first receiving activity.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. The units shall be prepared for mobile shipment. When sufficient space is available, packaged and unpackaged components and sections shall be placed and secured on the basic unit in any available space that will not interfere with the lifting or mobility of the unit. The components shall be secured to the unit with zinc-coated strapping conforming to QQ-S-781 or QQ-S-790. Packaged and unpackaged components and sections that cannot be secured to the basic unit shall be packed in crates conforming to PPP-C-650, style A, type as applicable, or MIL-C-3774, type I, style A. Blocking, bracing and anchoring shall be in accordance with the applicable crate specification and the appendix to MIL-C-104. Crated components shall be shipped at the same time as the basic unit.

5.3.2 Level C. The plant, components, repair parts, tools, and publications shall be prepared for shipment in compliance with Uniform Freight Classification Rules or with other carrier rules applicable to the mode of transportation.

5.4 Marking. Packages, shipping containers, and mobile plants shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The crushing and screening plant is intended for use in the production of aggregate for roadways, airstrips, parking aprons, and similar projects.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When an initial production plant is not required (see 3.3).
(c) Number of plants required for initial production inspection (see 3.3).
(d) Whether the dolly shall be compatible with the military or commercial dump truck (see 3.13.2).
(e) Technical publications required (see 3.23).
(f) Applicable list of repair parts and maintenance tools (see 3.24).
(g) Time interval for submission of electromagnetic compatibility reports (see 4.6.3.6 and 6.5).
(h) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).
(i) Sectionalization required (see 5.2.1.1).

6.3 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the crushing and screening plant covered by this specification as contained in the Federal Supply Catalog, DOD Section, Identification list C9100-IL for FSC Group 91.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.27.

6.5 Electromagnetic compatibility. The contracting officer should submit a copy of the report required by 4.6.3.6 to the Director, U.S. Army Electronics Laboratories, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey.

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6.6 Revision changes. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - ME
Air Force - 84

Review activities:

Army - EL
Air Force - 84

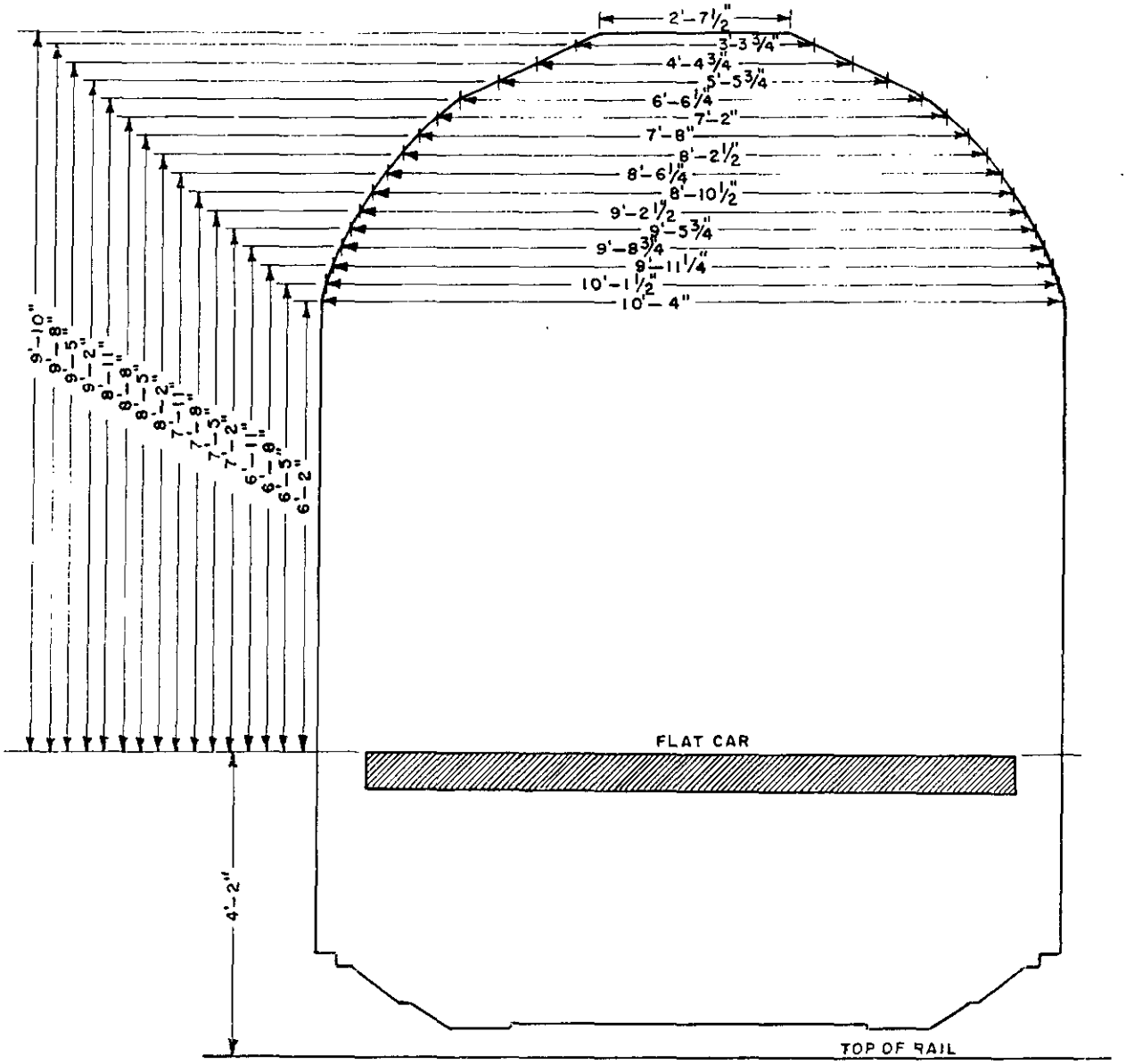
Preparing activity:

Army - ME

Project No. 3820-0093

Code C

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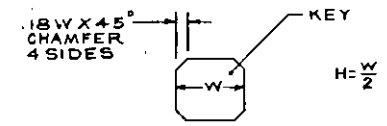


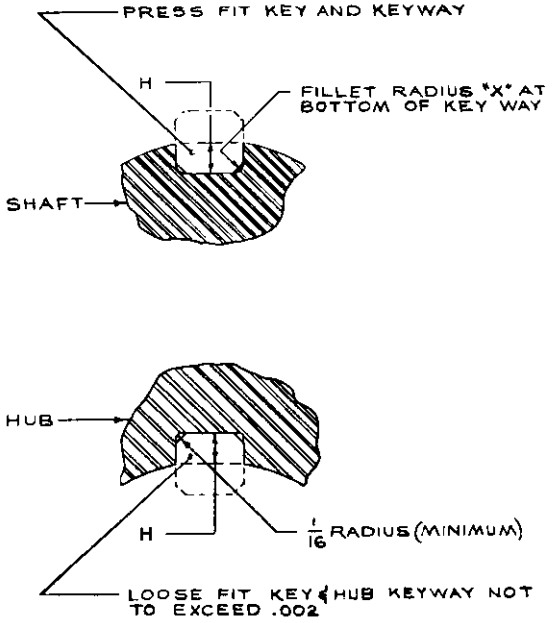
ITEM LENGTH-32'-10"

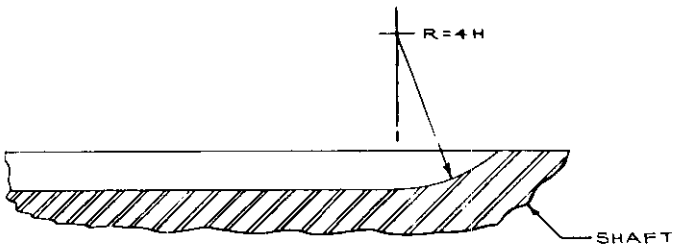
Figure 1 BERNE INTERNATIONAL
(CLEARANCE DIAGRAM)

A13123-1

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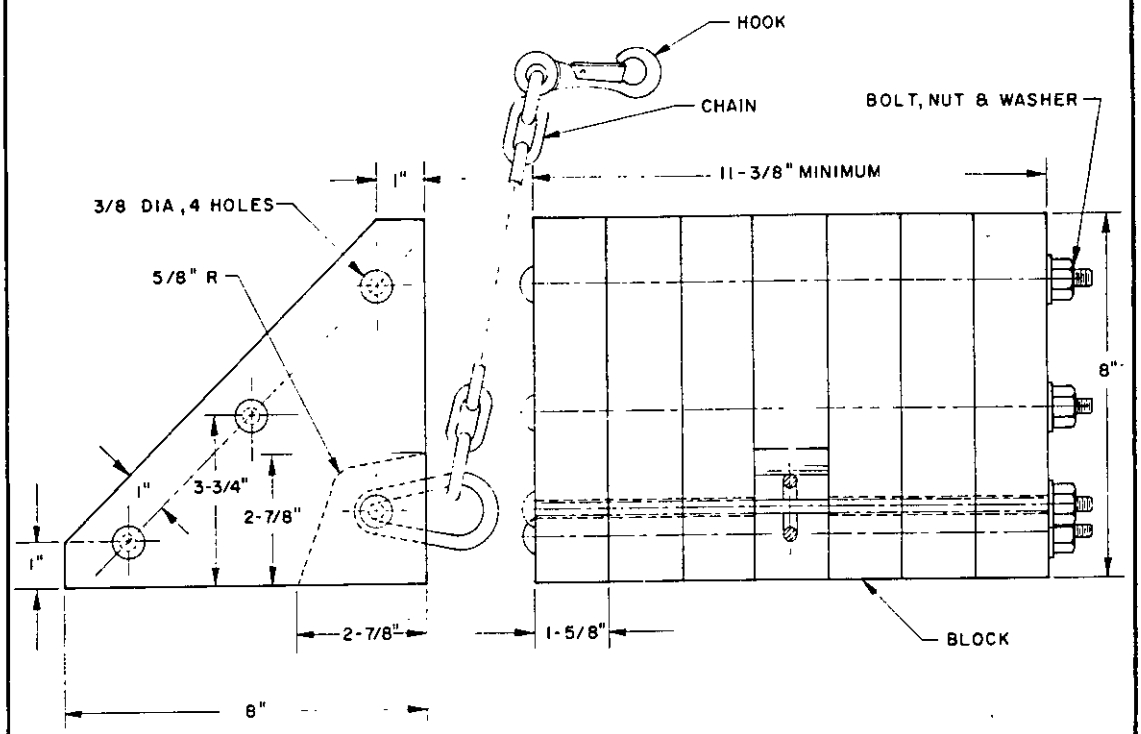
TYPICAL KEYWAY CONTOUR

FILLET RADIUS AT BOTTOM OF KEYWAY	
H	X
UNDER 1/4	SAE
1/4 TO 5/16	3/32
3/8	1/8
7/16 TO 1/2	5/32
9/16	3/16
5/8 TO 11/16	7/32
3/4	1/4
13/16 TO 7/8	9/32
15/16	5/16
1 TO 1 1/16	11/32
1 1/8 TO 1 1/2	3/8

FIGURE 2. KEYS AND KEYWAYS

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CHAIN DETAILS

CHAIN WELDED, CLOSE STRAIGHT LINK, GALV'D, 1/4" NOMINAL SIZE OF CHAIN BARS.

LENGTH OF 100 LINKS = 100 INCHES

MINIMUM TENSILE STRENGTH = 3,400 LBS.

WEIGHT / 100 FT. = 70 LBS.

MATERIAL NOTE

WOOD PARTS, FABRICATED, HARDWOOD.

FIGURE 3. CHOCK BLOCK ASSEMBLY

DX-603

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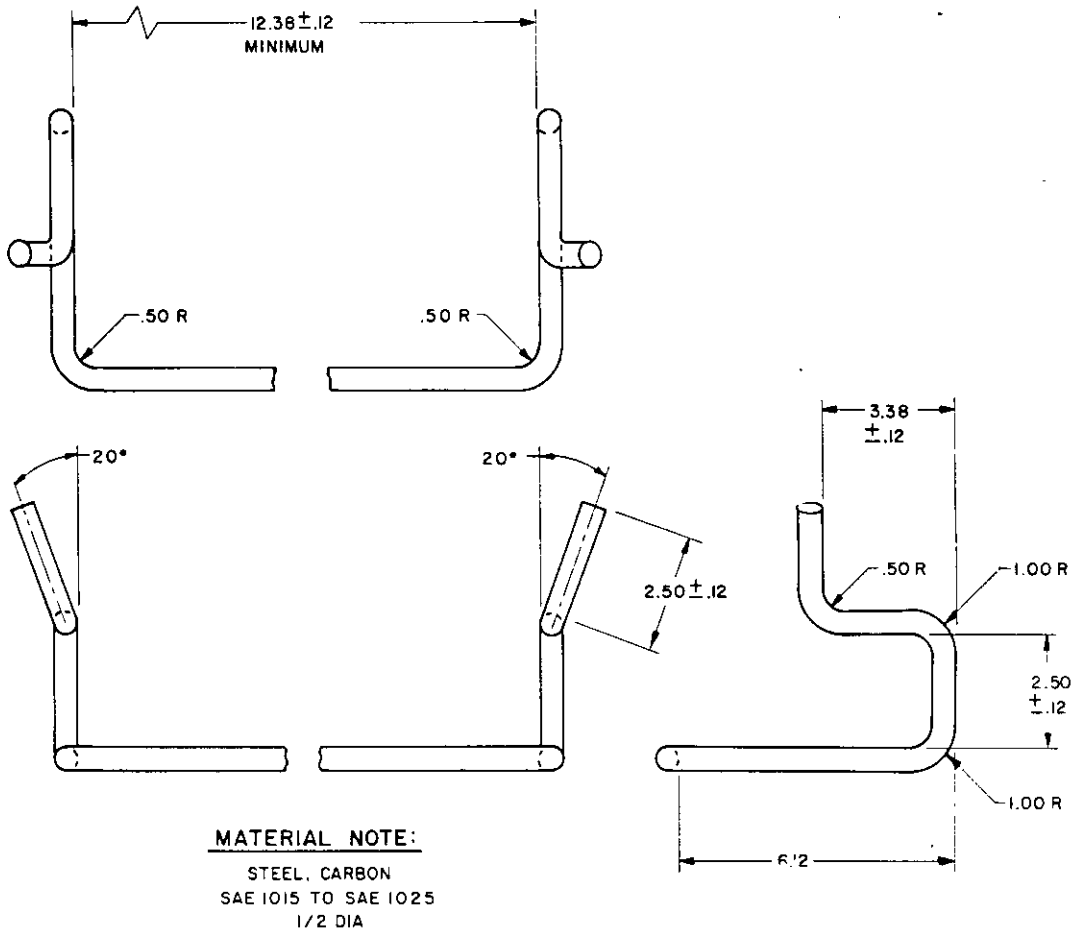


FIGURE 4. CRADLE

DX-604

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